

CENTRE FOR ADVANCED STUDIES

**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH,
LUCKNOW**



EVALUATION SCHEME AND SYLLABI

For

B.Tech.2nd Year
Computer Science & Engineering
Computer Science & Engineering (AI/ML)

Effective from the Session: 2026-27

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH.
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DETAILED SYLLABI

B.Tech. Second Year

Effective-from Session 2026-27

B.Tech. (Computer Science & Engineering) Second Year, Semester-III

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 301	Discrete Mathematics	T	BS	3	0	0	20	10	30	70	100	3
2.	CCS 301	Software Engineering	T	ES	3	0	0	20	10	30	70	100	3
3.	CCS 302	Design & Analysis of Algorithms	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 303	Object Oriented Programming	T	ES	3	0	0	20	10	30	70	100	3
5.	CCS 304	Database Management Systems	T	ES	3	1	0	20	10	30	70	100	4
6.	CCS 351	Software Engineering Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CCS 353	Object Oriented Programming Lab	P	ES	0	0	2	-	50	50	50	100	1
8	CCS 354	Database Management Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CVA 302	Universal Human Values*	P	VA	2*	0	0	20	10	30*	70*	100*	1
					15+	2	6			300+	500+70*	800+	21
10.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

VA: Value Added Course

B.Tech. (Computer Science & Engineering) Second Year, Semester-IV

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CCS 401	Computer Networks	T	ES	3	0	0	20	10	30	70	100	3
2.	CCS 402	Web Technology	T	ES	3	0	0	20	10	30	70	100	3
3.	CCS 403	Operating Systems	T	ES	3	0	0	20	10	30	70	100	3
4.	CCS 404	Theory of Computation	T	BS	3	1	0	20	10	30	70	100	4
5.	CCS 405	Probability and Statistics	T	ES	3	0	0	20	10	30	70	100	3
6.	CCS 451	Computer Networks Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CCS 452	Web Technology Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CCS 453	Operating Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CCS 456	Mini Project	P	ES	0	0	2	-	50	50	50	100	1
10.	CVA 401	Constitution of India*	T	VA	2*	0	0	20	10	30*	70*	100*	1
					15+2*	1	08			350+30*	550+70*	900+100*	21
11.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

PE: Program Elective

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course (4 week) during summer break after Semester-II and same will be assessed / evaluated in the Semester-III

B.Tech. Computer Science & Engineering (AI/ML) Second Year, Semester-III

										Evaluation Scheme			
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+T A	TE/PE	SW+ES E	Cr
1.	CAS 301	Discrete Mathematics	T	BS	3	0	0	20	10	30	70	100	3
2.	CAI 301	Software Engineering	T	ES	3	0	0	20	10	30	70	100	3
3.	CAI 302	Design & Analysis of Algorithms	T	ES	3	1	0	20	10	30	70	100	4
4.	CAI 303	Object Oriented Programming	T	ES	3	0	0	20	10	30	70	100	3
5.	CAI 304	Database Management Systems	T	ES	3	1	0	20	10	30	70	100	4
6.	CAI 351	Software Engineering Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CAI 353	Object Oriented Programming Lab	P	ES	0	0	2	-	50	50	50	100	1
8	CAI 354	Database Management Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CVA 302	Universal Human Values*	P	VA	2*	0	0	20	10	30*	70*	100*	1
					15+	2	6			300+	500+70*	800+	21
10.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

HS: Humanities and Social Science Course

VA: Value Added Course

B.Tech. Computer Science & Engineering (AI/ML) Second Year, Semester-IV

										Evaluation Scheme			
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAI 401	Computer Networks	T	ES	3	0	0	20	10	30	70	100	3
2.	CAI 402	Web Technology	T	ES	3	0	0	20	10	30	70	100	3
3.	CAI 403	Operating Systems	T	ES	3	0	0	20	10	30	70	100	3
4.	CAI 404	Theory of Computation	T	BS	3	1	0	20	10	30	70	100	4
5.	CAI 405	Probability and Statistics	T	ES	3	0	0	20	10	30	70	100	3
6.	CAI 451	Computer Networks Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CAI 452	Web Technology Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CAI 453	Operating Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CAI 456	Mini Project	P	ES	0	0	2	-	50	50	50	100	1
10.	CVA 401	Constitution of India*	T	VA	2*	0	0	20	10	30*	70*	100*	1
					15+2*	1	08			350+30*	550+70*	900+100*	21
11.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

PE: Program Elective

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course (4 week) during summer break after Semester-II and same will be assessed / evaluated in the Semester-III

B.Tech. (Computer Science & Engineering) and (Computer Science & Engineering (AI/ML)) Second Year, Semester-III

CAS 301: DISCRETE MATHEMATICS

Course Objectives:

The objective of this course is to provide students with foundational mathematical structures essential for computer science and engineering applications. The course aims to:

1. Develop mathematical maturity to understand and analyze discrete structures used in algorithms, data structures, and computer systems
2. Equip students with tools to model and solve computational problems using sets, logic, graphs, and counting principles.
3. Prepare students for advanced courses in algorithms, cryptography, networks, compilers, and artificial intelligence.
4. Enable students to apply discrete mathematical concepts to real-world computing problems.
5. Foster logical reasoning and proof-writing skills are crucial for theoretical computer science.

Content	Contact Hours
Unit-1:	8
Mathematical Logic: Statements and notations, Connectives, Well-formed formulas, Truth Tables, tautology, equivalence implication, Normal forms, Quantifiers, universal quantifiers. Predicates: Predicative logic, Free & Bound variables, Rules of inference, Consistency, proof of contradiction, Automatic Theorem Proving.	
Unit-2:	
Relations: Properties of Binary Relations, equivalence, transitive closure, compatibility and partial ordering relations, Lattices, Hasse diagram. Functions: Inverse Function Composition of functions, recursive Functions, Lattice and its Properties, Algebraic structures: Algebraic systems Examples and general properties, Semigroups and monads, groups sub groups' homomorphism, Isomorphism.	8
Unit-3:	
Elementary Combinatorics: Basis of counting, Combinations & Permutations, with repetitions, Constrained repetitions, Binomial Coefficients, Binomial Multinomial theorems, the principles of Inclusion – Exclusion. Pigeon hole principles and its application.	
Unit-4:	
Recurrence Relation: Generating Functions, Function of Sequences Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions. Characteristics roots solution of Inhomogeneous Recurrence Relation.	8
Unit-5:	
Graph Theory: Representation of Graph, DFS, BFS, Spanning Trees, planar Graphs. Graph Theory and Applications, Basic Concepts Isomorphism and Sub graphs, Multi graphs and Euler circuits, Hamiltonian graphs, Chromatic	

Numbers.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Apply concepts of sets, relations, functions, and propositional logic to solve problems related to discrete mathematical structures.	K2, K5
CO 2	Analyze binary relations, partial order relations, lattices, and algebraic structures such as semigroups, groups, and homomorphisms for modeling computational systems.	K1, K6 & K5
CO 3	Use principles of counting, permutations, combinations, inclusion–exclusion, and pigeonhole principle in solving combinatorial problems.	K3, K5
CO 4	Formulate and solve recurrence relations using substitution methods, characteristic roots, and generating functions.	K1, K4
CO 5	Apply graph theoretic concepts such as graph representation, traversal algorithms, spanning trees, Euler and Hamiltonian graphs in computer science and engineering applications.	K3, K4& K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books:

- Discrete Mathematics and its Applications, Kenneth H. Rosen, Fifth Edition. TMH.
- Discrete Mathematical structures Theory and application-Malik & Sen, Cengage.
- Discrete Mathematics with Applications, Thomas Koshy, Elsevier.
- Logic and Discrete Mathematics, Grass Man & Trembley, Pearson Education.

Reference Books:

- E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
- Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
- Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
- D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
- Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
- Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, McGraw-Hill; Sixth Edition.
- P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson Education.
- Advanced Engineering Mathematics. Chandrika Prasad, Reena Garg, 2018.
- H.K. Dass, Advanced Engineering Mathematics, H.K.Dass, S. Chand

CCS 301/CAI 301: Software Engineering

Course Objectives:

1. To understand the fundamentals of software development and SDLC models
2. To learn requirements analysis, system design, and modeling techniques
3. To gain knowledge of software testing, quality assurance, and maintenance
4. To introduce modern software development practices including Agile

Content	Contact Hours
Unit I: Introduction to Software Engineering	8
Software Engineering: Definition, Evolution, and Challenges; Software Crisis and its remedies; Software Development Life Cycle (SDLC); Process Models: Waterfall, Prototype, Spiral, Incremental; Agile Methodologies: Scrum, XP	
Unit II: Requirements Engineering	8
Requirements Engineering Process; Types of Requirements: Functional & Non-functional; Requirement Elicitation Techniques; Software Requirement Specification (SRS); Requirement Validation and Management	
Unit III: System Design	8
Design Concepts and Principles; Architectural Design: Layered, Client-Server; UML Diagrams: Use Case, Class, Sequence, Activity; Data Flow Diagrams (DFD); Cohesion and Coupling	
Unit IV: Software Testing & Quality Assurance	8
Testing Fundamentals and Strategies; Unit Testing, Integration Testing, System Testing; White-box and Black-box Testing; Software Quality Assurance (SQA); Reliability and Metrics	
Unit V: Software Project Management	8
Project Planning and Scheduling; Cost Estimation Techniques (COCOMO Model); Risk Management; Configuration Management; Software Maintenance and Evolution	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Understand software engineering principles and process models	K2
CO 2 Analyze and document software requirements	K3
CO 3 Design software systems using UML and architectural concepts	K4
CO 4 Apply testing strategies and quality assurance techniques	K4
CO 5 Manage software projects using estimation and planning tools	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Textbooks:

1. Software Engineering: A Practitioner's Approach – Roger S. Pressman
2. Software Engineering – Ian Sommerville

Reference Books:

1. Fundamentals of Software Engineering – Rajib Mall
2. Software Engineering – K.K. Aggarwal & Yogesh Singh

CCS 302/CAI 302: DESIGN & ANALYSIS OF ALGORITHMS

Course Objectives:

The objectives of this course are:

1. To understand algorithm design strategies and complexity analysis
2. To develop problem-solving skills using efficient algorithms
3. To analyze time and space complexity using asymptotic notations
4. To learn advanced algorithmic paradigms
5. To evaluate computational feasibility of problems

Content	Contact Hours
Unit-1	8
Introduction & Complexity Analysis: Definition and Characteristics of Algorithms; Analysis Framework; Asymptotic Notations: Big-O, Omega, Theta; Best, Average, Worst Case Analysis; Recurrence Relations and their solutions (Substitution, Recursion Tree, Master Method); Time-Space Trade-off.	
Unit-2	8
Divide and Conquer: General Method; Binary Search; Merge Sort; Quick Sort; Strassen's Matrix Multiplication; Analysis of Divide and Conquer Algorithms.	
Unit-3	8
Greedy Algorithms: Greedy Strategy; Properties of Greedy Method; Fractional Knapsack Problem; Job Sequencing with Deadlines; Minimum Spanning Tree (Kruskal's Algorithm and Prim's Algorithm); Huffman Coding.	
Unit-4	8
Dynamic Programming: Principle of Optimality; Difference between Greedy and Dynamic Programming; 0/1 Knapsack Problem; Longest Common Subsequence (LCS); Matrix Chain Multiplication; Optimal Binary Search Tree.	
Unit-5	8
Backtracking and Graph Algorithms: Backtracking: General Method; N-Queen Problem; Hamiltonian Cycle Problem; Graph Traversal: Breadth First Search (BFS), Depth First Search (DFS); Shortest Path Algorithms: Dijkstra's Algorithm; Introduction to NP-Completeness.	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Understand algorithm design principles and complexity analysis	K2
CO 2 Apply divide-and-conquer and greedy techniques	K3
CO 3 Design solutions using dynamic programming	K4
CO 4 Analyze graph algorithms and backtracking techniques	K4
CO5 Evaluate computational complexity and feasibility	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

TextBooks and Reference Books:

1. Introduction to Algorithms – Cormen, Leiserson, Rivest, Stein
2. Fundamentals of Algorithms – Ellis Horowitz & Sartaj Sahni
3. Algorithm Design – Kleinberg & Tardos
4. Design and Analysis of Algorithms – Aho, Hopcroft & Ullman

CCS 303/CAI 303: OBJECT ORIENTED PROGRAMMING

Course Objectives:

The objectives of this course are:

1. To understand the concepts of Object-Oriented Programming (OOP) and its advantages over procedural programming
2. To learn object-oriented concepts such as classes, objects, inheritance, polymorphism, and encapsulation
3. To develop programming skills using an object-oriented language (C++/Java)
4. To understand exception handling, file handling, and advanced OOP features
5. To design and implement real-world applications using object-oriented principles

Content	Contact Hours
UNIT I	8
Introduction to Object-Oriented Programming: Introduction to Programming Paradigms: Procedural vs Object-Oriented Programming; Basic Concepts of OOP: Objects, Classes, Encapsulation, Abstraction, Inheritance, Polymorphism; Benefits of OOP; Structure of C++/Java Program; Data Types, Variables, Operators; Control Structures: Conditional and Looping Statements.	
UNIT II	8
Classes and Objects: Class Definition and Objects; Access Specifiers (Public, Private, Protected); Member Functions; Constructors and Destructors; Types of Constructors; Function Overloading; Operator Overloading; Inline Functions; Friend Functions; Static Data Members and Member Functions.	
UNIT III	8
Inheritance and Polymorphism Inheritance: Types (Single, Multiple, Multilevel, Hierarchical, Hybrid); Access Control in Inheritance; Constructor and Destructor in Inheritance; Polymorphism: Compile-time and Run-time Polymorphism; Function Overriding; Virtual Functions; Abstract Classes; Dynamic Binding.	
UNIT IV	8
Exception Handling and File Handling: Exception Handling: Try, Catch, Throw Mechanism; Multiple Catch Blocks; User Defined Exceptions; File Handling: File Streams; Opening and Closing Files; File Modes; Reading and Writing Files; Binary and Text Files; Random Access to Files.	
UNIT V	8
Advanced OOP Concepts: Templates: Function Templates and Class Templates; Standard Template Library (STL) Basics (if using C++); Pointers and Dynamic Memory Allocation; Introduction to Smart Pointers (conceptual); Basics of Java Packages/Namespace; Introduction to GUI Programming (conceptual); Application Development using OOP Concepts.	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Understand the principles of object-oriented programming	K2
CO-2	Apply concepts of classes, objects, and inheritance	K3
CO-3	Analyze and implement polymorphism and dynamic binding	K4
CO-4	Apply exception handling and file handling techniques	K3
CO-5	Design real-world applications using advanced OOP concepts	K5

TEXT-BOOKS AND REFERENCE BOOKS:

1. E Balagurusamy, Object Oriented Programming with C++ | 8th Edition | McGraw Hill Education (India) Private Limited, Khasra No.: 305 to 311 & 320, NH - 91, Dadri bypass, Vill. Badhpura, Dadri, Dist. Gautam Budh Nagar, UP-203207, India; 18001035875
2. Schildt, H. (2023). Java: The Complete Reference, Twelfth Edition.
3. "C++ How to Program" by Harvey M. Deitel and Paul J. Deitel
4. Object-Oriented Programming in C++ – Robert Lafore

CCS 304/CAI 304: DATABASE MANAGEMENT SYSTEM

Course Objectives:

The objectives of this course are:

1. To understand the fundamental concepts of database systems
2. To learn database design using Entity-Relationship (ER) modeling
3. To gain knowledge of relational database concepts and SQL
4. To understand normalization and transaction management
5. To develop skills for designing and implementing database applications

Syllabus

Content	Contact Hours
Unit-1:Introduction to Database Systems	8
Introduction to Database Systems; File System vs Database System; Database System Concepts and Architecture; Data Models: Hierarchical, Network, Relational; DBMS Components; Data Independence; Database Users; Three Schema Architecture; Introduction to Database Languages.	
Unit-2:Entity-Relationship Model	8
Entity-Relationship (ER) Model; Entities, Attributes, Relationships; Types of Attributes; Keys: Primary, Candidate, Super Key; ER Diagrams; Extended ER Model (Specialization, Generalization, Aggregation); Mapping ER Model to Relational Model.	
Unit-3:Relational Model and SQL	8
Relational Model Concepts; Relational Algebra; Tuple Relational Calculus (basic concepts); Structured Query Language (SQL): DDL, DML, DCL, TCL; Constraints: Primary Key, Foreign Key, Unique, Not Null; Joins, Subqueries, Views, Indexing.	
Unit-4:Database Design and Normalization	8
Functional Dependencies; Anomalies in Database; Normalization: 1NF, 2NF, 3NF, BCNF; Lossless Join and Dependency Preservation; Database Design Process; Introduction to Denormalization.	
Unit-5:Transaction Management and Advanced Topics	8
Transaction Concepts; ACID Properties; Concurrency Control; Lock-based Protocols; Deadlocks and Recovery Techniques; Database Security and Authorization; Introduction to NoSQL Databases and Distributed Databases.	

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)	Bloom's Level
CO1 Understand database concepts and architecture	K2
CO2 Design ER models and relational schemas	K3
CO3 Apply SQL for database manipulation and querying	K3
CO4 Analyze normalization and database design issues	K4
CO5 Evaluate transaction management and concurrency control techniques	K5

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Textbooks and Reference Books:

1. Database System Concepts – Abraham Silberschatz, Henry Korth, S. Sudarshan
2. Fundamentals of Database Systems – Elmasri & Navathe
3. Database Management Systems – Raghu Ramakrishnan
4. SQL The Complete Reference – Groff & Weinberg

CCS 351/CAI 351: SOFTWARE ENGINEERING LAB

Course Objectives:

The objectives of this lab are:

1. To provide practical exposure to software engineering concepts and methodologies
2. To understand requirement analysis and software design techniques
3. To develop skills in modeling using UML diagrams
4. To apply software testing and quality assurance techniques
5. To gain experience in software project development and documentation

List of Experiments (Indicative & not limited to)

1. Study of Software Development Life Cycle (SDLC) models through case studies
2. Preparation of Software Requirement Specification (SRS) for a given problem
3. Identification and documentation of functional and non-functional requirements
4. Design of Use Case Diagram for a real-world system
5. Design of Class Diagram and Object Diagram
6. Design of Sequence Diagram and Activity Diagram
7. Preparation of Data Flow Diagram (DFD) (Level 0, Level 1)
8. Design of software architecture (Layered / Client-Server)
9. Preparation of test cases for given software modules
10. Implementation of unit testing and integration testing concepts
11. Study and application of software metrics (basic measures)
12. Demonstration of version control system (e.g., Git basics)
13. Risk analysis and project planning for a software project
14. Mini Project: Development of a small software system including SRS, design, testing, and documentation

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	Understand software engineering processes and documentation	K2
CO2	Apply requirement analysis and UML modeling techniques	K3
CO3	Design software systems using standard design methodologies	K4
CO4	Apply testing techniques and quality assurance practices	K3
CO5	Develop and document a complete software project	K5

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

CCS 353/CAI 353: OBJECT ORIENTED PROGRAMMING LAB

Course Objective:

The objectives of this lab are:

1. To provide practical exposure to object-oriented programming concepts
2. To develop programming skills using C++/Java
3. To implement classes, objects, inheritance, and polymorphism
4. To understand exception handling and file handling through programming
5. To design and develop small applications using object-oriented principles

List of Experiments (Indicative & not limited to)

1. Write a program to demonstrate basic structure of C++/Java program and input/output operations
2. Write a program to create a class and demonstrate the use of objects and member functions
3. Implement constructors and destructors in a class
4. Write programs to demonstrate function overloading and operator overloading
5. Implement use of friend functions and static data members
6. Write programs to demonstrate different types of inheritance (single, multilevel, multiple, hierarchical)
7. Implement function overriding and runtime polymorphism using virtual functions
8. Write a program to demonstrate abstract class and pure virtual functions
9. Implement exception handling using try, catch, and throw blocks
10. Write programs for file handling: reading and writing text and binary files
11. Implement dynamic memory allocation using pointers
12. Write programs using templates (function templates and class templates)
13. Develop applications using OOP concepts (e.g., Student Management System, Library System)
14. Mini Project: Design and implement a real-world application using OOP principles

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	Understand basic object-oriented programming concepts through practice	K2
CO2	Apply classes, objects, and inheritance in program development	K3
CO3	Implement polymorphism and dynamic binding concepts	K3
CO4	Apply exception handling and file handling techniques	K3
CO5	Develop real-world applications using object-oriented principles	K5

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

CCS 354/CAI 354: DATABASE MANAGEMENT SYSTEMS LAB

Course Objectives:

The objectives of this lab are:

1. To provide hands-on experience in database design and implementation
2. To develop skills in SQL queries and database operations
3. To understand normalization and relational database design
4. To implement transactions and constraints in databases
5. To build simple database-driven applications

List of Experiments (Indicative & not limited to)

1. Study of DBMS environment and installation of database software (MySQL/Oracle/PostgreSQL)
2. Creation of database and tables using SQL (DDL commands)
3. Perform basic SQL operations: INSERT, UPDATE, DELETE (DML commands)
4. Implement constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL
5. Perform queries using SELECT with WHERE, ORDER BY, GROUP BY, HAVING
6. Implementation of different types of joins: INNER JOIN, OUTER JOIN, SELF JOIN
7. Use of aggregate functions: COUNT, SUM, AVG, MAX, MIN
8. Nested queries and subqueries
9. Creation and use of Views
10. Implementation of Indexes
11. Database design using ER diagrams for a real-world application
12. Normalization of database tables (up to 3NF)
13. Implementation of transactions (COMMIT, ROLLBACK, SAVEPOINT)
14. Introduction to stored procedures and triggers (basic)
15. Mini project: Design and implement a database system (e.g., Library, Student Management, Hospital System)

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	Understand database creation and SQL commands	K2
CO2	Apply SQL queries for data manipulation	K3
CO3	Design relational database schemas	K4
CO4	Implement constraints, joins, and transactions	K3
CO5	Develop a database-driven application	K5

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

CVA 302: UNIVERSAL HUMAN VALUES

Course Objectives:

The objectives of this course are:

1. To help students understand the importance of human values in personal and professional life
2. To develop a holistic perspective towards life, society, and nature
3. To enable students to differentiate between values and skills, and understand their role in harmony
4. To promote ethical and responsible behavior in society
5. To encourage self-exploration and continuous self-improvement

Syllabus

Content	Contact Hours
Unit-1:Introduction to Human Values	8
Need, Basic Guidelines, Content and Process for Value Education; Self-Exploration: What is it? Its content and process; 'Natural Acceptance' and Experiential Validation; Continuous Happiness and Prosperity; Right understanding, Relationship and Physical Facility; Understanding the Human Being as Co-existence of Self (I) and Body.	
Unit-2:Understanding Harmony in the Human Being	8
Harmony in the Human Being: Understanding the needs of Self (I) and Body; Understanding the Body as an Instrument of 'I'; Harmony in 'I' (Self): Thoughts, Behavior, Work; Harmony of Self with Body; Programs to ensure self-regulation and health.	
Unit-3:Understanding Harmony in the Family and Society	8
Harmony in Family: Trust and Respect as the foundational values; Other feelings: Affection, Care, Guidance, Reverence, Glory, Gratitude, Love; Harmony in Society: Understanding values in human relationships; Justice in human-to-human relationships; Role of family in society.	
Unit-4:Understanding Harmony in Nature and Existence	8
Harmony in Nature: Interconnectedness and mutual fulfillment among four orders of nature (Material, Plant, Animal, Human); Understanding existence as co-existence; Environmental sustainability; Role of human beings in maintaining harmony with nature.	
Unit-5:Implications of Human Values in Professional Ethics	8
Professional Ethics: Natural acceptance of human values; Ethical human conduct; Competence in professional ethics; Holistic technologies, production systems and management models; Case studies on ethical practices; Social responsibility and sustainable development.	

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)	Bloom's Level
CO1 Understand the significance of human values and self-exploration	K2
CO2 Analyze harmony within self and between self and body	K4
CO3 Apply values to improve relationships in family and society	K3

CO4	Understand harmony in nature and environmental sustainability	K2
CO5	Apply ethical principles in professional and social life	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Textbooks

- A Foundation Course in Human Values and Professional Ethics – R.R. Gaur, R. Sangal, G.P. Bagaria

Reference Books:

- Human Values – A.N. Tripathi
- Professional Ethics and Human Values – M. Govindarajan, S. Natarajan, V.S. Senthil Kumar

**B.Tech. (Computer Science & Engineering) and (Computer Science & Engineering)
(AI/ML) Second Year, Semester-IV**

CCS 401/CAI 401: COMPUTER NETWORKS

Course Objectives:

The objectives of this course are:

1. To understand the fundamental concepts of computer networks and communication systems
2. To learn different network models and architectures
3. To study data transmission, error detection, and flow control mechanisms
4. To understand routing, switching, and congestion control techniques
5. To gain knowledge of modern networking technologies and security issues

Syllabus:

Content	Contact Hours
Unit-1: Introduction to Computer Networks	8
Introductory Concepts: Goals and applications of networks, Categories of networks, Organization of the Internet, ISP, Network structure and architecture (layering principles, services, protocols and standards), The OSI reference model, TCP/IP protocol suite, Network devices and components. Physical Layer: Network topology design, Types of connections, Transmission media, Signal transmission and encoding, Network performance and transmission impairments, Switching techniques and multiplexing.	
Unit-2: Data Link Layer: Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols). Medium Access Control and Local Area Networks: Channel allocation, Multiple access protocols, LAN standards, Link layer switches & bridges (learning bridge and spanning tree algorithms).	8
Unit-3: Network Layer Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.	
Unit-4: Transport Layer: Process-to-process delivery, Transport layer protocols (UDP and TCP), Multiplexing, Connection management, Flow control and retransmission, Window management, TCP Congestion control, Quality of service.	8
Unit-5: Application layer Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login, Network management,	

Data compression, Cryptography – basic concepts.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission	K1, K2
CO 2	Apply channel allocation, framing, error and flow control techniques.	K3
CO 3	Describe the functions of Network Layer i.e. Logical addressing, subnetting & Routing Mechanism.	K2, K3
CO 4	Explain the different Transport Layer function i.e. Port addressing, Connection Management, Error control and Flow control mechanism.	K2, K3
CO 5	Explain the functions offered by session and presentation layer and their Implementation.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Textbooks:

1. Behrouz Forouzan, "Data Communication and Networking", McGraw Hill
2. Andrew Tanenbaum "Computer Networks", Prentice Hall.
3. William Stallings, "Data and Computer Communication", Pearson.
4. Kurose and Ross, "Computer Networking- A Top-Down Approach", Pearson.
5. Peterson and Davie, "Computer Networks: A Systems Approach", Morgan Kaufmann
6. W. A. Shay, "Understanding Communications and Networks", Cengage Learning.
7. D. Comer, "Computer Networks and Internets", Pearson.
8. Behrouz Forouzan, "TCP/IP Protocol Suite", McGraw Hill.

CAS 402/CAI 402: WEB TECHNOLOGY

Course Objectives

The objective of this course is to help students understand the fundamentals of web application development, covering both client-side and server-side technologies. The course aims to:

1. Introduce the basic concepts, architecture, and protocols of the World Wide Web.
2. Develop proficiency in building dynamic and responsive web pages using HTML, CSS, and JavaScript.
3. Provide knowledge of server-side scripting for dynamic content generation and data management.
4. Familiarize students with contemporary web development frameworks and architectures such as Node.js, Express, and RESTful APIs.
5. Enable students to integrate databases, web services, and cloud platforms for full-stack web development.

Syllabus:

Content	Contact Hours
Unit-1: Introduction to Web Technologies and HTML5	
Introduction to Internet and World Wide Web, Client-Server Architecture, Web Browsers and Web Servers, HTTP and HTTPS protocols, Static and Dynamic Websites. HTML5 Fundamentals: Structure of HTML documents, Formatting tags, Hyperlinks, Images, Tables, Lists, Forms and Input Controls, Semantic HTML5 tags, Audio and Video elements, Canvas, Local Storage and Session Storage. CSS3 Fundamentals: Selectors, Box Model, Flexbox, CSS Grid, Responsive Web Design, Media Queries, Bootstrap framework.	8
Unit 2: JavaScript and Client-Side Scripting	
Introduction to JavaScript: Variables, Data Types, Operators, Expressions, Control Structures, Functions, Arrays and Objects, Scope of Variables. Modern JavaScript (ES6+): Arrow Functions, Template Literals, Destructuring, Modules. Document Object Model (DOM): Event Handling, Form Validation, Dynamic DOM Manipulation. AJAX and Fetch API: JSON, Asynchronous Requests, Error Handling, API Integration. Simple interactive web applications using JavaScript.	8
Unit-3: React.js	
Introduction to React.js, Virtual DOM, JSX Syntax, Components and Props, Functional and Class Components, State and Lifecycle Methods. React Hooks: use State, use Effect, use Context. Forms and Event Handling in React, Conditional Rendering, Lists and Keys, React Router. API Integration in React Applications, State Management Basics, Project Structure and Build Tools. Development of Single Page Applications (SPA) using React.js.	
Unit-4: Angular JS	
Introduction to Angular JS, MVC Architecture, Modules and Controllers, Expressions and Filters. Directives and Data Binding, Dependency Injection,	8

Services and Factories. Routing in Angular JS, Form Handling and Validation, HTTP Services. Single Page Application development using Angular JS. Comparison between React.js and Angular JS frameworks.	
Unit-5: Next.js and Full Stack Development	
Introduction to Next.js, File-Based Routing, Pages and Layouts. Server Side Rendering (SSR), Static Site Generation (SSG), Dynamic Routing.API Routes and Backend Integration, Authentication Basics, Environment Variables. Image Optimization and Performance Enhancement. Deployment using Vercel and modern hosting platforms. Building full-stack applications using Next.js.	8

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Design responsive web pages using HTML5 and CSS3.Implement client-side functionality using JavaScript and DOM manipulation.	K2
CO 2	Develop single-page applications using React.js.Analyze and implement MVC architecture using Angular JS	K3
CO 3	Create full-stack applications using Next.js.Integrate frontend applications with REST APIs and databases.	K3, K4
CO 4	Implement server-side solutions using Node.js, Express, and databases for complete web applications.	K4, K5
CO 5	Apply authentication, routing, and state management concepts in web applications. Deploy modern web applications on cloud platforms.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

TEXTBOOKS:

- Web Technologies — Uttam K. Roy, Oxford University Press.
- JavaScript: The Definitive Guide — David Flanagan, O'Reilly.
- Learning React — Alex Banks & Eve Porcello.
- Pro AngularJS — Adam Freeman.
- Learning Next.js — Michele Riva.

REFERENCE BOOKS:

- Beginning Web Programming — Jon Duckett, WROX.
- Programming the World Wide Web — Robert W. Sebesta, Pearson.
- Eloquent JavaScript — Marijn Haverbeke.
- Full Stack React — Anthony Accomazzo.
- AngularJS Up and Running — Brad Green & Shyam Seshadri.
- Fullstack React with Next.js — Newline Publications.

CAS 403/CAI 403: OPERATING SYSTEMS

Course Objectives

The main objective of this course is to provide a comprehensive understanding of modern operating system concepts, structures, and algorithms used in process management, memory organization, file systems, and I/O management.

The course aims to:

1. Introduce the fundamental components and structure of operating systems.
2. Explain various process management, scheduling, and synchronization techniques.
3. Address memory management, paging, segmentation, and virtual memory mechanisms.
4. Explore file storage organization, I/O systems, and protection features.
5. Prepare students to compare different operating systems and analyze real-world OS design issues.

Syllabus:

Content	Contact Hours
Unit-1: Operating System Fundamentals & Structures	
Introduction to Operating Systems: Objectives, functions, evolution, types of operating systems (batch, multiprogramming, multitasking, time-sharing, real-time, distributed, mobile, cloud OS). System Components: Kernel, user interface, command interpreters, system calls, system booting process. OS Structure: Layered approach, microkernel, modular approach, Monolithic kernels. OS Services: System calls, API, Command Line Interface vs GUI, OS design and implementation issues.	8
Unit-2: Process and CPU Scheduling	
Processes: Concept, process states, PCB (Process Control Block), operations on processes, context switching. Threads: Multithreading models, advantages, and issues. Process Scheduling: Scheduling criteria, Pre-emptive & Non-Pre-emptive Scheduling, and algorithms (FCFS, SJF, RR, Priority Scheduling, Multilevel Queue Scheduling). Inter-Process Communication (IPC): Shared memory, message passing, synchronization. Case Study: Process management in Linux/Windows.	8
Unit-3: Process Synchronization and Deadlocks	
Process Synchronization: Critical section problem, hardware synchronization, Semaphores, Monitors, Classical synchronization problems (Producer-Consumer, Dining Philosophers, Readers-Writers). Deadlocks: Definition, necessary conditions, Resource Allocation Graph (RAG), Deadlock prevention, avoidance (Banker's Algorithm), detection, and recovery methods.	
Unit-4: Memory Management and Virtual Memory	
Memory Management: Concept of contiguous and non-contiguous allocation, fixed and variable partitioning, fragmentation (internal & external). Paging & Segmentation: Basic concepts, page tables, page replacement algorithms (FIFO, LRU, Optimal), and effective access time. Virtual Memory: Concept, demand paging, Thrashing, working set model. Allocation and Performance	8

Issues: Compaction, Swapping, Allocation of frames, and memory management in UNIX/Linux.	
Unit-5: File Systems, I/O, and Protection	
File System: File concepts, operations, access methods (sequential, indexed, direct), directory structure, file-sharing and protection. Implementation: File system interface and implementation, allocation methods (contiguous, linked, indexed), Free-space management, File system mounting, Journaling File Systems. I/O Systems: I/O hardware, device drivers, interrupt handling, DMA. Protection & Security: Goals, access matrix, authentication, encryption, threats, and security policies. Case Study: Linux/UNIX and Windows file systems.	8

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand fundamental concepts, structure, and services of operating systems.	K2
CO 2	Apply process management, scheduling, and synchronization algorithms to manage concurrent systems.	K3, K4
CO 3	Analyze and evaluate different memory management and virtual memory strategies.	K4, K5
CO 4	Examine the functioning of file systems and I/O management mechanisms.	K4, K5
CO 5	Evaluate protection and security mechanisms in modern operating systems.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

TEXT-BOOKS

1. Silberschatz, Galvin, Gagne, *Operating System Concepts*, 10th Edition, Wiley, 2020.
2. Andrew S. Tanenbaum, Herbert Bos, *Modern Operating Systems*, 4th Edition, Pearson Education, 2015.
3. William Stallings, *Operating Systems: Internals and Design Principles*, 9th Edition, Pearson Education, 2018.

REFERENCE BOOKS

1. D.M. Dhamdhare, *Operating Systems: A Concept-Based Approach*, 3rd Edition, McGraw-Hill Education.
2. Achyut S. Godbole, Atul Kahate, *Operating Systems*, McGraw-Hill Education, 2017.
3. Maurice J. Bach, *The Design of the UNIX Operating System*, Pearson Education, 2015.
4. Gary Nutt, *Operating Systems: A Modern Perspective*, Addison-Wesley, 2004.

CAS 404/CAI 404: THEORY OF COMPUTATION

Course Objectives

The objective of this course is to introduce the fundamental concepts and mathematical theory that form the foundation of computer science. The course emphasizes the models of computation, formal languages, and their computational limitations.

1. To understand the abstract machines and their capabilities in recognizing formal languages.
2. To develop the ability to construct finite automata, grammars, and other computational models.
3. To study the hierarchy of formal languages and their interrelationships.
4. To analyze decidability, computability, and the limits of algorithmic problem-solving.
5. To provide theoretical grounding necessary for compiler design, complexity analysis, and artificial intelligence.

Syllabus:

Content	Contact Hours
Unit-1: Introduction to Automata and Languages	
Introduction: Alphabets, Strings, Languages, Operations on Strings and Languages. Automata Theory Overview: Basic structure of finite automata, Transition functions, Acceptance of strings and languages. Finite Automata (FA): Deterministic Finite Automata (DFA), Non-Deterministic Finite Automata (NFA), Equivalence of DFA and NFA, NFA with ϵ -moves, Conversion from NFA to DFA. Applications: Finite automata for lexical analysis and pattern recognition.	8
Unit-2: Regular Languages and Grammars	
Regular Expressions (RE): Subset construction, Regular operations, Arden's theorem, Equivalence of RE and FA. Regular Grammars: Right and left linear grammars, Equivalence with finite automata. Properties of Regular Languages: Closure properties, Pumping lemma for regular languages, Decision properties, and Minimization of finite automata using partition method.	8
Unit-3: Context-Free Grammars (CFG) and Pushdown Automata (PDA)	
Context-Free Grammars: Definition, Derivation, Parse trees, Ambiguity, Simplification of CFG (elimination of ϵ -productions and unit productions), Normal forms (Chomsky and Greibach). Pushdown Automata (PDA): Definition, Representation, Computation by PDA, Conversion from CFG to PDA and vice versa, Acceptance by final state and empty stack, Comparison with FA.	
Unit-4: Turing Machines and Computability	
Turing Machines (TM): Definition, Representation, Transition diagram and table, Instantaneous descriptions, Design of TMs for simple tasks (copying, binary addition, palindrome checking). Variants of TM: Multi-track, Multi-tape and non-deterministic TMs, Equivalence of various TMs. Church-Turing Thesis: Linear Bounded Automata, Universal Turing Machine, Recursive and	8

Recursively Enumerable Languages, multi-tapetion of TM representations.	
Unit-5: Decidability, Undecidability, and Complexity	
Decidability: Decidable and undecidable problems, Halting problem, Post correspondence problem, Reducibility. Complexity Theory: Classes P, NP, NP-complete, NP-hard (basic introduction). Applications of Automata Theory: Compiler design (syntax analysis), Natural language processing, Verification systems, and Formal methods in software engineering.	8

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand the basic concepts of automata theory, formal languages, and their representations using abstract machines.	K2
CO 2	Design finite automata and regular expressions for lexical analysis and pattern recognition problems.	K3, K4
CO 3	Construct and analyze context-free grammars and pushdown automata for language parsing applications.	K3, K4
CO 4	Understand and analyze the concept of Turing machines and their role in defining algorithmic computation.	K4, K5
CO 5	Evaluate the decidability and computational complexity of problems and relate them to real-world computational limitations.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

TEXT-BOOKS

1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, *Introduction to Automata Theory, Languages and Computation*, 3rd Edition, Pearson Education, 2014.
2. Peter Linz, *An Introduction to Formal Languages and Automata*, 6th Edition, Jones and Bartlett Learning, 2016.
3. K.L.P. Mishra, N. Chandrasekaran, *Theory of Computer Science: Automata, Languages and Computation*, 3rd Edition, PHI Learning, 2006.

REFERENCE BOOKS

1. Harry R. Lewis, Christos H. Papadimitriou, *Elements of the Theory of Computation*, 2nd Edition, Pearson Education, 2006.
2. Martin J.C., *Introduction to Languages and the Theory of Computation*, 4th Edition, McGraw-Hill Education, 2011.
3. Michael Sipser, *Introduction to the Theory of Computation*, 3rd Edition, Cengage Learning, 2012.
4. Dexter C. Kozen, *Automata and Computability*, Springer, 2013.

CCS 405/CAI 405: PROBABILITY AND STATISTICS

Course Objectives:

The objectives of this course are:

1. To understand the fundamental concepts of probability theory
2. To learn random variables and probability distributions
3. To apply statistical methods for data analysis
4. To understand sampling theory and hypothesis testing
5. To develop analytical skills for solving real-world problems using probability and statistics

Syllabus:

Content	Contact Hours
Unit-1: Probability Theory	
Introduction to Probability; Sample Space and Events; Axioms of Probability; Conditional Probability; Bayes' Theorem; Independent Events; Random Experiments; Applications of Probability in Engineering Problems.	8
Unit-2: Random Variables and Probability Distributions	
Random Variables: Discrete and Continuous; Probability Mass Function (PMF) and Probability Density Function (PDF); Cumulative Distribution Function (CDF); Mathematical Expectation; Variance and Standard Deviation; Moments; Important Distributions: Binomial, Poisson, Normal Distribution.	8
Unit-3: Statistical Measures and Data Analysis	
Measures of Central Tendency: Mean, Median, Mode; Measures of Dispersion: Range, Variance, Standard Deviation; Correlation and Regression Analysis; Curve Fitting; Introduction to Time Series Analysis.	
Unit-4: Sampling and Estimation	
Sampling Techniques: Random Sampling, Stratified Sampling; Sampling Distribution; Standard Error; Estimation: Point and Interval Estimation; Confidence Intervals; Central Limit Theorem.	8
Unit-5: Hypothesis Testing	
Testing of Hypothesis; Null and Alternative Hypothesis; Type I and Type II Errors; Tests of Significance: Z-test, t-test, Chi-square test, F-test; Analysis of Variance (ANOVA); Applications in Engineering and Data Analysis.	8

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand probability concepts and laws	K2
CO 2	Apply probability distributions and random variables	K3
CO 3	Analyse statistical data using various measures	K4
CO 4	Apply sampling and estimation techniques	K3

CO 5	Evaluate hypotheses using statistical tests	K5
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K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Textbooks:

1. Probability and Statistics for Engineers and Scientists – Walpole, Myers, Myers, Ye
2. Introduction to Probability and Statistics – Sheldon M. Ross

Reference Books:

1. Probability and Statistics – S.C. Gupta & V.K. Kapoor
2. Schaum's Outline of Probability and Statistics – Murray R. Spiegel

CAS 451/CAI 451: COMPUTER NETWORKS LAB

Course Objectives

The main objective of this laboratory course is to provide students with practical exposure to the principles, operation, and configuration of modern computer networks. The course focuses on understanding network architectures, protocols, and performance analysis through experiments and simulations.

The objectives are to:

1. Enable students to understand basic networking concepts, transmission techniques, and protocol operations.
2. Provide hands-on experience in designing, implementing, and troubleshooting Local Area Networks (LANs) and Wide Area Networks (WANs).
3. Familiarize students with network simulation tools such as *Cisco Packet Tracer*, *NS2/NS3*, *Wireshark*, or *GNS3*.
4. Develop the ability to analyze network performance metrics like throughput, delay, loss, and utilization.
5. Prepare students for industrial networking roles involving configuration, monitoring, and securing networks.

List of Experiments (Indicative & Not Limited To)

1. Study of basic networking commands (ping, tracert, netstat, ipconfig, arp, nslookup, etc.) and their purpose in diagnostics.
2. Familiarization with network components—NICs, cables, switches, routers, and hubs. Creation of physical and logical topologies in Packet Tracer.
3. Implementation of IP addressing and subnetting for a given network scenario.
4. Understanding and configuring network topologies (Bus, Star, Ring, Mesh) using simulation tools.
5. Configuration of switches and routers for basic LAN and inter-LAN communication using Cisco Packet Tracer or GNS3.
6. Implementation of static and dynamic routing protocols: RIP, OSPF, and EIGRP and observation of routing tables.
7. Simulation of data flow using Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) using NS2/NS3.
8. Study of socket programming using C/Python for client-server communication (TCP and UDP).
9. Simulation of ARP, ICMP, and DNS operations using Wireshark—packet capture and protocol field analysis.
10. Implementation of error detection and correction techniques (CRC, checksum, parity bits).
11. Study and implementation of congestion control mechanisms such as Leaky Bucket and Token Bucket algorithms.
12. Implementation of VLANs (Virtual LANs) and Inter-VLAN routing using switch configurations.
13. Study of firewall configuration and packet filtering using iptables/Windows Defender Firewall.
14. Setup of basic WLAN configuration and testing connectivity through Wi-Fi adapters.
15. Mini project: Design and simulation of an enterprise network integrating routing, switching, and security mechanisms.

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Demonstrate understanding of fundamental networking concepts, protocols, and topologies through hands-on practice..	K2
CO 2	Configure routers, switches, and end devices for establishing LAN/WAN communication	K3, K4
CO 3	Analyze and simulate packet transmission using network simulation tools to understand protocol behavior.	K4, K5
CO 4	Develop socket-based programs for real-time client-server communication and packet handling.	K4, K5
CO 5	Design, test, and troubleshoot small to medium-scale computer networks using routing and switching principles.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text-Books

1. Andrew S. Tanenbaum, David J. Wetherall, *Computer Networks*, 5th Edition, Pearson Education, 2017.
2. Behrouz A. Forouzan, *Data Communications and Networking*, 5th Edition, McGraw-Hill Education, 2020.
3. James F. Kurose, Keith W. Ross, *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson Education, 2021.

Reference Books

1. William Stallings, *Data and Computer Communications*, 10th Edition, Pearson Education, 2016.
2. Rajesh Maurya, *Network Simulation with Cisco Packet Tracer*, McGraw-Hill, 2021.
3. Larry L. Peterson, Bruce S. Davie, *Computer Networks: A Systems Approach*, 6th Edition, Morgan Kaufmann, 2021.
4. S. Thiruvengadam, *Networks Lab Manual*, Khanna Publishing House, 2022.

CAS 452/CAI 452: WEB TECHNOLOGY LAB

Course Objectives

The primary objective of this laboratory course is to provide students with practical experience in designing and developing dynamic and interactive web applications. The course focuses on both **client-side** and **server-side** technologies using modern web development tools and frameworks.

The course aims to:

1. Familiarize students with the fundamental concepts of front-end web development using HTML, CSS, and JavaScript.
2. Enable students to design, implement, and deploy responsive and dynamic web pages.
3. Introduce server-side scripting using Node.js, Express, or similar frameworks for back-end logic.
4. Develop database connectivity and data-driven web applications.
5. Encourage integration of full-stack components into a complete web-based project with best practices for security and optimization.

List of Experiments (Indicative & Not Limited To)

1. Study of different HTML tags and creation of web pages using HTML5 (text, image, forms, lists, tables).
2. Develop a personal or portfolio webpage with hyperlinks, multimedia elements, and navigation bars.
3. Apply external CSS3 styles to design ergonomic and responsive layouts using **Flexbox** and **Grid systems**.
4. Implement JavaScript programs for form validation, image slider/gallery, and DOM manipulation.
5. Use JavaScript **ES6 features** such as arrow functions, classes, and promises in a small project.
6. Develop a small dynamic web page using **Bootstrap** or **Tailwind CSS** for responsive design.
7. Implement **AJAX / Fetch API** for asynchronous data retrieval from a server.
8. Create a **Node.js and Express.js** web server and build RESTful API endpoints for CRUD operations.
9. Perform database connectivity using **MongoDB or MySQL**, implementing data insertion, update, and retrieval functionalities.
10. Develop a simple user registration and login system with validation and session/cookie management.
11. Implement client-server communication using **JSON** data exchange and error handling.
12. Design a simple **single-page application (SPA)** using front-end frameworks like **React.js** or **Vue.js** (optional).
13. Integrate third-party APIs (e.g., Weather API, Currency Converter API) into a web application.
14. Deploy the developed web application on a local server or cloud (e.g., **Firebase, Vercel, or Netlify**).
15. **Mini Project:** Full-stack web application demonstrating front-end design, server integration, database connectivity, and deployment.

Course Outcomes:

Course Outcome (CO)	Bloom's Level
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At the end of this course, the students will be able to:		
CO 1	Demonstrate proficiency in front-end web development using HTML, CSS, and JavaScript.	K2, K3
CO 2	Develop responsive, interactive, and structured web interfaces using CSS frameworks and client-side scripting.	K3, K4
CO 3	Implement and test server-side logic using Node.js/Express.js and integrate with databases.	K4, K5
CO 4	Apply asynchronous communication and web service integration using RESTful APIs.	K3
CO 5	Design and deploy a complete full-stack dynamic web application with security and accessibility features.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text-books

1. Uttam K. Roy, *Web Technologies*, Oxford University Press, 2018.
2. Achyut S. Godbole, Atul Kahate, *Web Technologies – TCP/IP to Internet Application*, McGraw-Hill Education, 2020.
3. Paul Deitel & Harvey Deitel, *Internet and World Wide Web: How to Program*, 6th Edition, Pearson Education, 2019.

Reference Books

1. Robin Nixon, *Learning PHP, MySQL & JavaScript with CSS & HTML5*, O'Reilly, 2021.
2. Ethan Brown, *Web Development with Node and Express*, 2nd Edition, O'Reilly, 2019.
3. Chris Northwood, *Full Stack Web Development: With React and Node*, Apress, 2021.

CAS 453/CAI 453: OPERATING SYSTEMS LAB

Course Objectives

The primary objective of this laboratory course is to complement the theoretical concepts learned in the Operating Systems course by providing hands-on experience in process management, memory handling, and file system operations.

The course aims to:

1. Develop practical understanding of how an operating system manages processes, resources, and memory.
2. Implement algorithms for CPU scheduling, process synchronization, and deadlock handling.
3. Learn interprocess communication and system call mechanisms in Linux/Unix environments.
4. Explore basic shell scripting and file-handling operations.
5. Enable students to analyze system performance and design small-scale OS simulations.

List of Experiments (Indicative & Not Limited To)

1. Study of basic Linux/Unix commands and their use in OS-level operations.
2. Write a shell script to display system configuration details (OS type, version, CPU info, memory usage, etc.).
3. Implementation of file handling operations using system calls: `open()`, `read()`, `write()`, `lseek()`, and `close()`.
4. Program to demonstrate the creation and termination of processes using system calls like `fork()`, `exec()`, and `wait()`.
5. Simulation of CPU scheduling algorithms: FCFS, SJF, Round Robin, and Priority Scheduling — compare turnaround and waiting times.
6. Implement the concept of Inter-Process Communication (IPC) using Pipes, FIFOs, or Message Queues.
7. Solve the **Producer–Consumer** problem using semaphores in C using `pthread` library.
8. Solve the **Reader–Writer problem** and **Dining Philosophers problem** using monitors or semaphores.
9. Implement Banker’s Algorithm for deadlock avoidance.
10. Program to simulate page replacement algorithms: FIFO, LRU, and Optimal.
11. Implement memory allocation strategies – First Fit, Best Fit, and Worst Fit.
12. Write a shell script for automating batch file processing (e.g., backup, search, or compression).
13. Simulate disk scheduling algorithms such as FCFS, SSTF, SCAN, and C-SCAN.
14. Implement file allocation methods – Contiguous, Linked, and Indexed.
15. Mini Project: Simulation of a simplified Operating System module covering process scheduling or memory management features.

Course Outcomes:

Course Outcome (CO)		Bloom’s Level
At the end of this course, the students will be able to:		
CO 1	Demonstrate proficiency in executing system-level operations and understanding Linux system calls.	K2, K3
CO 2	Implement CPU scheduling and process synchronization algorithms	K3, K4

	for efficient resource allocation.	
CO 3	Analyze and simulate different process, memory, and file management techniques.	K4, K5
CO 4	Construct shell scripts to automate OS tasks and file operations.	K3, K6
CO 5	Design and evaluate simplified OS models to analyse performance and behaviour.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text-Books

1. Silberschatz, Galvin, Gagne, *Operating System Concepts*, 10th Edition, Wiley, 2020.
2. Andrew S. Tanenbaum, *Modern Operating Systems*, 4th Edition, Pearson Education, 2015.
3. William Stallings, *Operating Systems: Internals and Design Principles*, 9th Edition, Pearson Education, 2018.

Reference Books

1. D.M. Dhamdhere, *Operating Systems: A Concept-Based Approach*, McGraw-Hill Education, 2017.
2. Sumitabha Das, *Your UNIX – The Ultimate Guide*, Tata McGraw-Hill, 2016.
3. Maurice J. Bach, *Design of the UNIX Operating System*, Pearson Education, 2015.
4. Albert S. Woodhull and Andrew S. Tanenbaum, *Operating Systems Design and Implementation*, Pearson, 2006.

CCS 456/CAI 456: MINI PROJECT (01 Credit)

Course Objectives:

The student is required to undertake a short project based on the syllabus. The objective is to provide hands-on training in the practical aspects of the curriculum and to enhance their research and analytical skills.

CVA-401 CONSTITUTION OF INDIA (Mandatory — 1 Credit)

Course Objectives

The objective of this course is to familiarize students with the fundamental principles of the Indian Constitution and its significance in democratic governance.

The students will learn:

- Understand the historical background, making and salient features of the Indian Constitution.
- Study Fundamental Rights, Fundamental Duties and Directive Principles of State Policy.
- Understand the structure and functioning of the Parliamentary and Federal systems of India.
- Study Constitutional amendments, Emergency provisions and Local Self Government schemes.
- Appreciate the role of the Constitution in shaping Indian democracy and governance.

Syllabus

Unit 1: Introduction and Basic Information about Indian Constitution	Contact Hours
Meaning of the constitution law and constitutionalism, Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Enforcement of the Constitution, Indian Constitution and its Salient Features, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government – Constitutional Scheme in India.	8
Unit 2: Union Executive and State Executive	
Powers of Indian Parliament Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister, Judiciary – The Independence of the Supreme Court, Appointment of Judges, Judicial Review, Public Interest Litigation, Judicial Activism, Lokpal, Lok-Ayukta, The Lokpal and Lok-Ayuktas Act 2013, State Executives – Powers and Functions of the Governor, Powers and Functions of the Chief Minister, Functions of State Cabinet, Functions of State Legislature, Functions of High Court and Subordinate Courts.	8
Unit 3: Introduction and Basic Information about Legal System	
The Legal System: Sources of Law and the Court Structure: Enacted law -Acts of Parliament are of primary legislation, Common Law or Case law, Principles taken from decisions of judges constitute binding legal rules. The Court System in India and Foreign Courtiers (District Court, District Consumer Forum, Tribunals, High Courts, Supreme Court). Arbitration: As an alternative to resolving disputes in the normal courts, parties who are in dispute can agree that this will instead be referred to arbitration. Contract law, Tort, Law at workplace.	8
Unit 4: Intellectual Property Laws and Regulation to Information	

Intellectual Property Laws- Introduction, Legal Aspects of Patents, Filing of Patent Applications, Rights from Patents, Infringement of Patents, Copyright and its Ownership, Infringement of Copyright, Civil Remedies for Infringement, Regulation to Information- Introduction, Right to Information Act, 2005, Information Technology Act, 2000, Electronic Governance, Secure Electronic Records and Digital Signatures, Digital Signature Certificates, Cyber Regulations Appellate Tribunal, Offences, Limitations of the Information Technology Act.	8
Unit 5: Business Organizations and E-Governance	
Sole Traders, Partnerships: Companies: The Company's Act: Introduction, Formation of a Company, Memorandum of Association, Articles of Association, Prospectus, Shares, Directors, General Meetings and Proceedings, Auditor, Winding up. E-Governance and role of engineers in E-Governance, Need for reformed engineering serving at the Union and State level, Role of I.T. professionals in Judiciary.	8

Course Outcomes:

	Course Outcome (CO)	Bloom's Level
CO1	Understand the historical background, making, and salient features of the Indian Constitution including Fundamental Rights and Duties.	K2
CO2	Analyze the structure and functions of the Union and State Executives, and the role of the Judiciary in Indian governance.	K4
CO3	Understand the legal system, sources of law, court structure, and alternative dispute resolution mechanisms.	K2
CO4	Understand intellectual property laws and regulations governing information technology and digital governance	K2
CO5	Apply knowledge of business organizations and e-governance to understand the role of engineers in democratic society.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books

1. Brij Kishore Sharma; Introduction to the Indian Constitution, PHI, New Delhi, latest edition.
2. Granville Austin; The Indian Constitution: Corner stone of a Nation. 1966, Oxford Clarendon Press.
3. Subhash C. Kashyap; Our Constitution: An Introduction to India's Constitution and 35 constitutional Law, NBT, 2018.
4. P. M. Bakshi; The Constitution of India, Latest Edition, Universal Law Publishing.

Reference Books:

1. V. K. Ahuja; Law Relating to Intellectual Property Rights (2007)
2. Suresh T. Viswanathan: The Indian Cyber Laws, Bharat Law House, New Delhi-88
3. P. Narayan; Intellectual Property Law, Eastern Law House, New Delhi